

From Joint Variables to the

Geometric Jacobian

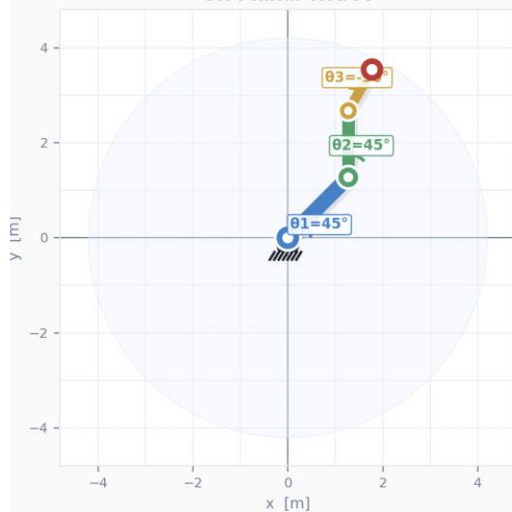
Fundamental velocity relationship:

$$\dot{\xi} = J(q) \dot{q}$$

Slide 1 — Configuration Vector q

1 / 5

3R Planar Robot



Concept

The robot's configuration is fully described by the joint angle vector:

$$q = [\theta_1 \ \theta_2 \ \theta_3]^T \in \mathbb{R}^3$$

Each angle θ_i is measured relative to the previous link. Drag the sliders to explore how q maps to different arm shapes.

Current configuration q

$$\theta_1 = +45.0^\circ \quad \theta_2 = +45.0^\circ \quad \theta_3 = -30.0^\circ$$

$$p_{EE} = (+1.773, +3.539) \text{ m}$$

Jacobian $J_v(q) \in \mathbb{R}^{2 \times 3}$

	Col 1 (joint 1)	Col 2 (joint 2)	Col 3 (joint 3)
v_x	-3.539	-2.266	-0.866
v_y	+1.773	+0.500	+0.500

Manipulability $\mu = \sqrt{\det(J_v J_v^T)}$

2.3659

✗ full rank ✓

full rank

Joint angles (drag to explore):

θ_1 0.785

θ_2 0.785

θ_3 -0.524

← Prev

Next →

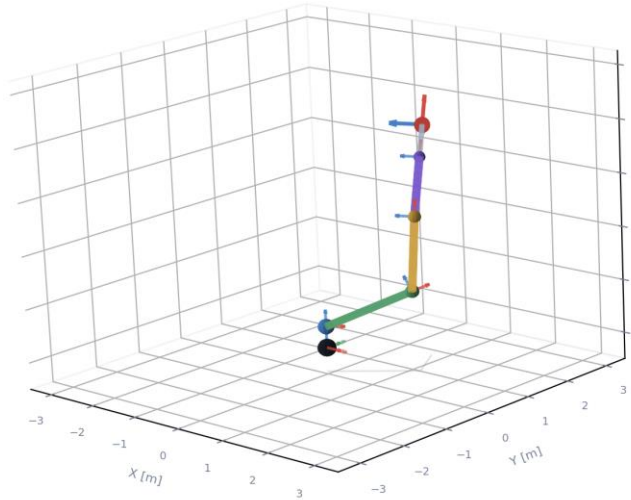
Reset

← Prev / Next → to cycle through lecture slides

Slide 1 – Configuration $q \in \mathbb{R}^4$

1 / 5

4R Spatial Robot



Homogeneous Transformation $T(q) \in SE(3)$

4x4 matrix = rotation R + translation p + projective row

col 1	col 2	col 3	col 4
-0.2501	-0.5343	-0.8074	+0.8259
+0.4364	+0.6822	-0.5866	+1.6507
$R \in SO(3)$ orientation +0.8643	-0.4990 $p = (+0.826, +1.651, +3.924) \text{ m}$	+0.0626	+3.9238
+0.0000	+0.0000	+0.0000	+1.0000

$p \in \mathbb{R}^3$
position

Geometric Jacobian $J(q) \in \mathbb{R}^{6 \times 4}$

Upper 3 rows: linear J_v Lower 3 rows: angular J_ω

	j1	j2	j3	j4
v_x	-1.651	+2.851	+2.319	-0.908
v_y	+0.826	+2.071	+1.685	+1.160
v_z	+0.000	-1.638	-0.180	-0.848
ω_x	+0.000	-0.588	-0.588	-0.807
ω_y	+0.000	+0.809	+0.809	-0.587
ω_z	+1.000	+0.000	+0.000	+0.063

$\mu = 10.1649$ ✓ full rank

Concept

A 4-DOF spatial robot is described by:

$$q = [\theta_1 \ \theta_2 \ \theta_3 \ \theta_4]^T \in \mathbb{R}^4$$

Joint types (all revolute):

- θ_1 – base yaw (rotates around Z)
- θ_2 – shoulder (rotates around Y)
- θ_3 – elbow (rotates around Y)
- θ_4 – wrist roll (rotates around Z)

The colored frames show each joint's local coordinate system.

← Prev

Next →

Reset

What Is the Configuration Vector q ?

The robot configuration is fully described by a vector of joint variables:

$$q = [q_1 \ q_2 \ \cdots \ q_n]^T \in \mathbb{R}^n$$

Each component represents one degree of freedom:

Revolute Joint

$$q_i = \theta_i \quad (\text{joint angle})$$

Angle θ_i measured at the joint axis

Prismatic Joint

$$q_i = d_i \quad (\text{joint displacement})$$

Linear displacement d_i along the axis

Examples: 2R manipulator $q = [\theta_1 \ \theta_2]^T$ — 6-DOF robot $q = [\theta_1 \cdots \theta_6]^T$

Forward Kinematics Describes Pose

From previous lectures, the pose of a robot is a homogeneous transformation:

Pose mapping

$$T(q) \in SE(3)$$

where

$$T(q) = \begin{bmatrix} R(q) & p(q) \\ 0_{1 \times 3} & 1 \end{bmatrix}$$

Components

R(q)

Rotation matrix $\in SO(3)$

encodes end-effector orientation

p(q)

Position vector $\in \mathbb{R}^3$

encodes end-effector position

T(q) tells us WHERE the end-effector is — but not HOW FAST it moves.

From Position to Velocity

Key question in robotics control:

If the joints move with velocities $\dot{q}$, what is the velocity of the end-effector [in manipulators]?

What we have — Pose

Position:

$$p = p(q)$$

Orientation:

$$R(q) \in SO(3)$$

Captured by:

$$T(q) \in SE(3)$$

What we need — Velocity

Linear velocity:

$$v = \dot{p} \in \mathbb{R}^3$$

Angular velocity:

$$\omega \in \mathbb{R}^3$$

Combined:

$$\xi = [v \ \omega]^T \in \mathbb{R}^6$$

Linear Velocity and the Jacobian $J_v(q)$

Derivation

1 From forward kinematics:

$$p = p(q)$$

2 Differentiate with respect to time:

$$\dot{p} = \frac{\partial p}{\partial q} \dot{q}$$

3 Linear velocity relationship:

$$v = J_v(q) \dot{q}$$

Definition

$$J_v(q) = \frac{\partial p}{\partial q} \in \mathbb{R}^{3 \times n}$$

The linear Jacobian $J_v(q)$ is the matrix of partial derivatives of the end-effector position with respect to joint variables.

Properties

- Each column j describes how joint j 's motion affects v
- Computed analytically from forward kinematics
- Depends on current configuration q

Angular Velocity and the Jacobian $J\omega(q)$

Similarly, the orientation $R(q)$ depends on joint variables. Differentiating: $S(\omega) = \dot{R}R^T$

Orientation from FK

$$R = R(q)$$

Time derivative

$$\dot{R} = \frac{\partial R}{\partial q} \dot{q}$$

Angular velocity relationship

$$\omega = J_{\omega}(q) \dot{q}$$

Why use ω instead of $\dot{R}$ directly?

Overcomplete representation

$\dot{R}$ has 9 entries but rotation has only 3 DOF — redundant and harder to work with.

Physical meaning

$\omega \in \mathbb{R}^3$ is the instantaneous axis-angle rotation rate — compact and intuitive.

Consistent with Newton–Euler

Angular velocity ω appears naturally in rigid-body dynamics and torque equations.

The Twist — Combining Linear and Angular Velocity

The instantaneous motion of a rigid body requires both linear and angular velocity.

The twist vector ξ (xi) is defined as:

$$\xi = \begin{bmatrix} v \\ \omega \end{bmatrix} \in \mathbb{R}^6$$

Components

$$v \in \mathbb{R}^3$$

Linear velocity

Rate of change of end-effector position: $v = \dot{p}$

$$\omega \in \mathbb{R}^3$$

Angular velocity

Instantaneous rotation rate of the end-effector frame

Constructing the Geometric Jacobian $J(q)$

Since both v and ω are linear in $\dot{q}$, we can combine them into a single matrix equation:

Step 1 — write both relationships

$$v = J_v(q) \dot{q}$$

$$\omega = J_\omega(q) \dot{q}$$

Step 2 — stack into block form

$$\xi = \begin{bmatrix} J_v(q) \\ J_\omega(q) \end{bmatrix} \dot{q}$$

Define the Geometric Jacobian

$$J(q) = \begin{bmatrix} J_v(q) \\ J_\omega(q) \end{bmatrix}$$

and obtain the fundamental relationship:

$$\xi = J(q) \dot{q}$$

Interpretation of the Geometric Jacobian

$J(q)$ provides a linear mapping between joint velocities and end-effector velocity:



Important: this mapping is valid instantaneously at the current configuration q . $J(q)$ changes as the robot moves.

Dimensions of the Geometric Jacobian

For an n-joint manipulator:

$$J(q) \in \mathbb{R}^{6 \times n}$$

because $\xi \in \mathbb{R}^6$ (6 rows) and $\dot{q} \in \mathbb{R}^n$ (n columns)

Special cases

n = 6

Square

$J \in \mathbb{R}^{6 \times 6}$ — invertible when non-singular. Standard for industrial arms.

n < 6

Deficient

$J \in \mathbb{R}^{6 \times n}$ with $n < 6$ — cannot independently control all 6 task DOF.

n > 6

Redundant

$J \in \mathbb{R}^{6 \times n}$ with $n > 6$ — infinite solutions. Extra DOF for optimization.

Structure of $J(q)$

	$\dot{q}_1$	$\dot{q}_2$	...	$\dot{q}_n$	
v_x					J_v
v_y					
v_z					
ω_x					J_ω
ω_y					
ω_z					

Why the Geometric Jacobian Is Important

The relationship $\xi = J(q) \dot{q}$ enables a wide range of robotics capabilities:

Compute end-effector velocity

Given $\dot{q}$, compute $\xi = J(q) \dot{q}$ directly.

$$\dot{\xi} = J(q) \dot{q}$$

Detect kinematic singularities

Singular when $\det(J) = 0$ — loss of DOF.

$$\det(J(q)) = 0$$

Design velocity controllers

Cartesian-space control via joint velocity inputs.

Cartesian motion planning

Track end-effector paths in task space.

Force/torque control

Joint torques from Cartesian force: $\tau = J^T F$

$$\tau = J^T(q) F$$

Inverse differential kinematics

Solve for $\dot{q}$ given a desired twist ξ .

$$\dot{q} = J^+(q) \xi$$

Summary

Key results from this lecture:

$$q \in \mathbb{R}^n$$

fully describes robot configuration (joint angles or displacements)

$$T(q) \in SE(3)$$

forward kinematics — gives pose, not velocity

$$\xi = [v^T \ \omega^T]^T \in \mathbb{R}^6$$

the twist — compact representation of end-effector velocity

$$J(q) \in \mathbb{R}^{6 \times n}$$

the geometric Jacobian — maps joint to Cartesian velocities

$$\xi = J(q) \dot{q}$$

fundamental velocity relationship — central result of this lecture